ESTIMATING THE IMPACT OF THE HAJJ: RELIGION AND TOLERANCE IN ISLAM'S GLOBAL GATHERING*

DAVID CLINGINGSMITH
ASIM IJAZ KHWAJA
MICHAEL KREMER

We estimate the impact on pilgrims of performing the Hajj pilgrimage to Mecca. Our method compares successful and unsuccessful applicants in a lottery used by Pakistan to allocate Hajj visas. Pilgrim accounts stress that the Hajj leads to a feeling of unity with fellow Muslims, but outsiders have sometimes feared that this could be accompanied by antipathy toward non-Muslims. We find that participation in the Hajj increases observance of global Islamic practices, such as prayer and fasting, while decreasing participation in localized practices and beliefs, such as the use of amulets and dowry. It increases belief in equality and harmony among ethnic groups and Islamic sects and leads to more favorable attitudes toward women, including greater acceptance of female education and employment. Increased unity within the Islamic world is not accompanied by antipathy toward non-Muslims. Instead, Hajjis show increased belief in peace, and in equality and harmony among adherents of different religions. The evidence suggests that these changes are likely due to exposure to and interaction with Hajjis from around the world, rather than to a changed social role of pilgrims upon return.

I. INTRODUCTION

We take advantage of a lottery used by Pakistan to allocate visas for the Hajj pilgrimage to understand the impact of the Hajj

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on pilgrims' attitudes, beliefs, and practices. Our work sheds light on contemporary concerns regarding Islamic orthodoxy and extremism by showing that although the Hajj increases observance of orthodox Islamic practices, it also increases the desire for peace and tolerance toward others, both Muslims and non-Muslims. Our study also contributes to a broader literature on theories of social interaction and social identity and on the role of religious institutions.

During five specific days of each year, more than two million Muslim men and women from over one hundred different countries gather in Mecca for the Hajj, often staying for over a month. Pilgrims mix across the lines of ethnicity, nationality, sect, and gender that divide them in everyday life. They affirm a common identity by communally performing identical rituals and dressing in similar garments that emphasize their equality.

Numerous pilgrim accounts suggest that the Hajj inspires feelings of unity with the worldwide Muslim community (Wolfe 1997). Malcolm X performed the Hajj after breaking from the heterodox Nation of Islam to become a Sunni Muslim. In a letter from Mecca, he wrote, "There were tens of thousands of pilgrims, from all over the world. . . . We were all participating in the same ritual, displaying a spirit of unity and brotherhood that my experiences in America had led me to believe never could exist between the white and non-white. . . . [W]hat I have seen, and experienced, has forced me to rearrange much of my thought-patterns previously held, and to toss aside some of my previous conclusions" (X 1965, p. 346).

Some have worried, though, that by promoting greater unity among Muslims, the Hajj could have negative implications for non-Muslims. After it emerged that some of the July 7 bombers of the London public transport system had undertaken the Hajj, the British intelligence services began monitoring pilgrims (*Sunday Times* 2007). Historically, colonial authorities also expressed similar concerns regarding the Hajj (Bose 2006; Low 2007).

Others have expressed concern that the Hajj promotes a particular type of Islam. For example, Naipaul (1981) laments what he sees as the erosion of local religious traditions in South Asian Islam in favor of a more Saudi or Arab version of Islam.

Of course, it is difficult to isolate the causal impact of the Hajj based on examples such as those of Malcolm X or the July 7 bombers. Those who choose to undertake the Hajj differ from those who do not, and the choice to do so may reflect other life changes. Thus, changes in pilgrims' views and behavior after the Hajj may

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not reflect its impact. We estimate the effect of performing the Hajj by comparing successful and unsuccessful applicants to a lottery Pakistan uses to allocate its limited supply of Hajj visas. Because our survey included 1,600 Pakistani Sunni Hajj visa applicants and was conducted five to eight months after the completion of the Hajj, our results should be interpreted as isolating the medium-term impact of performing the Hajj on this particular population.

Our results support the idea that the Hajj helps to integrate the Muslim world, leading to a strengthening of global Islamic beliefs, a weakened attachment to local religious customs, and a sense of unity and equality with others who are ordinarily separated in everyday life by sect, ethnicity, nationality, or gender, but who are brought together during the Hajj. Although the Hajj may help forge a common Islamic identity, there is no evidence that this is defined in opposition to non-Muslims. On the contrary, the notions of equality and harmony appear to extend to adherents of other religions as well. These results contrast sharply with the view that increased Islamic orthodoxy goes hand in hand with extremism.

We find that Hajjis (those who have performed the Hajj) are more likely to undertake universally accepted global Muslim religious practices such as fasting and performing obligatory and supererogatory (optional) prayers. In contrast, the Hajj reduces performance of less universally accepted, more localized practices and beliefs such as using amulets and the necessity of giving dowry. For example, the Hajj increases regular prayer in the mosque by 26% and almost doubles the likelihood of nonobligatory fasting. At the same time, it reduces the practice of using amulets by 8% and the South Asian belief according lower marriage priority to widows than to unmarried women by 18%.

The evidence suggests that the Hajj increases tolerance both within the Islamic world and also beyond it. Hajjis return with more positive views toward people from other countries. Hajjis are also more likely to state that various Pakistani ethnic and Muslim sectarian groups are equal, and that it is possible for such groups to live in harmony. These views of equality and harmony extend to non-Muslims as well: Hajjis are 22% more likely to declare that people of different religions are equal and 11% more likely to state that adherents of different religions can live in harmony.

We also find evidence that Hajjis are more peacefully inclined. For example, although few in our sample are willing to condemn the goals of Osama Bin Laden openly, Hajjis are almost twice as

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likely to do so. Hajjis are also more likely to express a preference for peace with India and are 17% more likely to declare that it is incorrect to physically punish someone who has dishonored the family.

There is little evidence that participating in the Hajj increases support for an increased role of religion in the state or politics, or that it induces negative views of the West. Hajjis are in fact less likely to believe that the state should enforce religious injunctions and that religious leaders should be able to dispense justice. Hajjis and non-Hajjis report similar views regarding the adoption of Western values and on the plausibility of Western/Jewish roles in the September 11 and July 7 terrorist attacks.

The feelings of unity and equality brought about by the Hajj extend across gender lines to an extent. Hajjis report more positive views on women's attributes and abilities. For example, they are 6% more likely to think women are spiritually better than men, an increase of over 50%. They also express greater concern about women's quality of life in Pakistan relative to other countries and about crimes against women in Pakistan. Hajjis are also more likely to support girls' education and female participation in the professional workforce. Hajjis show an 8% increase in their declared preference for their daughters or granddaughters to adopt professional careers. Male Hajjis show changes in views similar to those of female Hajjis. However, not all views on gender change. In particular, Hajjis are no more likely to question Islamic doctrine, such as unequal inheritance laws across gender, or to express views that potentially challenge male authority within the household, such as the correctness of a woman divorcing her husband. This suggests that Pakistani Hajjis' altered views on women reflect a movement away from local prejudices against women and toward fairer treatment within Islam, rather than a more general trend toward feminism.

Hajjis, primarily women, report lower levels of emotional and physical well-being. This may be due to the physically taxing nature of the Hajj rituals, as well as changed beliefs and greater awareness of the Muslim world outside Pakistan, particularly for women.

Contrary to some of the historical literature on the Hajj (cf. Azarya [1978]; Donnan [1989]; Yamba [1995]), we do not find evidence in our sample of major changes, at least in the medium term, in the social role or engagement of Hajjis after their return.

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Although our study cannot definitively determine what drives the impact of the Hajj, further evidence suggests that the results are not driven by changes in pilgrims' social roles upon return but rather reflect changes the pilgrim experiences during the Hajj, particularly exposure to Muslims from around the world. Hajjis gain experiential knowledge of the diversity of Islamic practices and beliefs, gender roles within Islam, and, more broadly, the world beyond Pakistan. Although the Hajj effects could be driven by a change in religious commitment, we do not find that Hajjis acquire greater formal religious knowledge. The Hajj's impact on experiential knowledge and on some tolerant attitudes toward other groups tends to be larger for those traveling in smaller groups, who are more likely to have a broad range of social interactions with people from different backgrounds during the Hajj. Hajjis also show the largest positive gain in their views of other nationalities in relation to Indonesians, the non-Saudi group they are most likely to observe during the Hajj. Hajjis' changed views toward women may also reflect an exposure channel, because the Hajj offers Pakistani pilgrims a novel opportunity to interact with members of the opposite gender in a religious setting, and to observe cross-gender interactions among Muslims from nations more accepting of such interactions.

Our results shed light on contemporary concerns about Islamic orthodoxy and extremism. For many in the West the link is apparent: 45% of Americans believe Islam is more likely to encourage violence than other religions and close to one-third use negative words such as fanatic, radical, and terror to describe their impressions of Islam (PEW Forum 2007). It is noteworthy that although the Hajj leads to greater religious orthodoxy, it also increases pilgrims' desire for peace and tolerance toward others, both Muslims and non-Muslims.

Our results also connect to a broad, longstanding literature on social interaction and the shaping of beliefs and identity. Laboratory experiments suggest that group interactions exacerbate conflict in competitive settings and lessen it in cooperative ones (DeVries and Slavin 1978; Stephan 1978; Johnson and Johnson 1983; Aronson and Patnoe 1997; Slavin and Cooper 1999; Pettigrew and Tropp 2006). Drawing on a particular real-world setting, Boisjoly et al. (2006) report evidence that exposure to African-American roommates generates more positive attitudes toward African-Americans among white students. In contrast, Fisman et al. (2008) find that exposure to a different race in youth makes

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individuals less likely to prefer that race for a potential mate as adults. Although social identity theory suggests that strengthening attachment to an in-group may lead to negative feelings toward an out-group (Sherif et al. 1954; Tajfel 1970; Tajfel and Turner 1986), our evidence shows that Hajjis also positively update their views both toward groups to which they were exposed to and those they were not.

Our findings also relate to a question in the sociology and economic modeling of religion about why religions often incorporate individually costly practices, and more broadly about the impact of religion on development (Iannacone 1992; Glaeser and Glendon 1998; Berman 2000; Sacerdote and Glaeser 2001; Barro and McCleary 2003; Guiso, Sapienza, and Zingales 2003). Putnam (2007) suggests that, in the U.S. context, religion may play a particularly important role as a "glue" that builds social capital. Our results suggest that the Hajj may play a role in contributing to the survival of Islam as a unified world religion. Over time, religions with far-flung adherents tend to evolve separate strands. Absent a central hierarchy in Islam, the Hajj may help bind the Islamic world together by moving Hajjis toward a common set of practices, making them more tolerant of others, and by creating a stronger shared identity.

Further work would be needed to determine the extent to which our findings generalize beyond the specific context we examine. Our survey, conducted five to eight months after the Hajj, captures medium-term effects. Although it remains open whether these effects persist, we find few changes over the survey period. Further examination is also needed to determine the extent to which the results generalize beyond our sample of Pakistani Hajj lottery applicants. For example, the impact of the Hajj on gender attitudes may be smaller for pilgrims from countries with more liberal gender views. To the extent that the gender results reflect a convergence to the mean views of Hajjis, the Hajj may even induce more conservative views for pilgrims from these countries. Moreover, nearly half of lottery applicants are illiterate. Although this is fairly representative of Pakistan and a large number of Muslim countries, results could differ in more educated societies. Finally, we assess the impact of the Hajj using survey questions, several of which elicit self-reported beliefs and opinions. Additional work is needed to determine how the Hajj would impact pilgrims' actions.

The remainder of this paper is organized as follows. Section II gives some background on the Hajj, focusing on aspects that

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contextualize our findings. Section III lays out our statistical approach, outlines aspects of the visa application process that are important for our identification strategy, and gives details of the survey. Section IV presents the main empirical results on religious practice and belief, tolerance, gender, and well-being. Section V explores potential channels for the observed effects. Section VI concludes with some underpinnings and broader implications of our results.

II. The Hajj Experience

The pilgrimage to Mecca is one of the five pillars of Islam and is obligatory for those with sufficient financial means. Many Hajjis describe it as the most significant religious event in their lives. Although the Hajj rituals last five days, many pilgrims stay longer. Most of the Hajjis in our sample report spending 40 days worshipping in the cities of Mecca and Medina.[1]

The Hajj is an inherently communal experience, in a religion that gives particular importance to communal rituals (McCleary 2007). Each ritual component of the Hajj is performed simultaneously with over two million participants. The focus is on individual practice rather than building religious knowledge. Moreover, each participant's performance is believed to reinforce the others', providing a shared aspect to individual worship.

The Hajj engenders substantial mixing across national, sectarian, and gender lines in an atmosphere that emphasizes equality and unity. Pilgrims' common identity is affirmed through common dress—a simple white garment known as the *ihram*—and the communal performance of standardized ritual practices.[2] Men shave their heads, making them more similar in physical appearance, and those who complete the Hajj are entitled to use the honorific Hajji/Hajjin as a prefix to their name, further emphasizing their common identity.

Although opportunities for in-depth intergroup interactions are limited both by language barriers and by the housing of pilgrims with their compatriots, the group nature of the experience makes observations of the contrasting practices and social dynamics of other groups even more salient. Close to two-thirds of the

1. See the Appendix for a timeline of the Hajj and an outline of the rituals and activities.

2. Men wear two white sheets. Women face less stringent requirements but typically also wear white.

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Hajjis in our survey reported interacting with people from other countries frequently during the Hajj.

The Hajj also involves more gender mixing than is typical among the Pakistani pilgrims we study. In Pakistan, interaction between men and women who are strangers is uncommon. Women rarely go to the mosque and when they do, they typically pray in a separate area from men. With equal numbers of male and female Hajjis (Bianchi 2004), such gender interactions are a natural part of the Hajj. Parties of pilgrims stay and move together for ease of planning and safety, and often include non–family members. Men pray alongside women, both Pakistani and non-Pakistani, during the Hajj. Our qualitative interviews revealed that these experiences were both very salient and unusual for Hajjis, and that most viewed them positively.

Finally, Hajjis are also exposed to a degree of religious diversity within the recognized schools of thought, in a religiously sanctioned context in which all are accepted. The fourteenth-century Muslim explorer Ibn Battuta noted that because followers of different schools of Islam prayed together at Mecca, this often led to mixing of religious practices (Ibn Battuta 2002 [1355]).

The Hajj is physically and financially taxing. Pilgrims travel over 80 km, much of it typically on foot. The extreme congestion heightens risks of injury and infectious disease (Ahmed, Arabi, and Memish 2006). Hajjis typically also sacrifice rest in order to maximize prayers during their stay. Participants in Pakistan's Hajj lottery system pay about US$2,000 each for the trip, roughly two and one-half times Pakistan's 2006 per capita GDP. The median respondent in our survey saved for the Hajj for over four years.

III. Methodology: The Hajj Lottery and Survey

Because those who choose to perform the Hajj are likely to be motivated by a wide spectrum of unobservable and potentially time-varying factors, such as religious commitment and the desire for spiritual transformation, it is difficult to measure the impact of the Hajj by comparing Hajjis and non-Hajjis. We address this by taking advantage of a lottery that allocates Hajj visas. Because successful and unsuccessful lottery applicants are *ex ante* identical in expectation, we can use the lottery outcome as an instrument for whether someone performs the Hajj and isolate the Hajj impact from potential confounding factors. This section provides details

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of the Hajj lottery, our statistical methodology, and our survey process.[3]

III.A. The Hajj Lottery Process

Historically, overcrowding on the Hajj has created logistical and safety problems. Saudi Arabia, on whose territory the Hajj takes place, therefore has established quotas for the number of Hajj visas available for each major Islamic country. For the January 2006 Hajj that we study, Pakistan's total quota was 150,000 visas. Ninety thousand visas were allocated by the government, the majority (89%) by randomized lottery and the remainder by special quotas for the military and civil service (Organization of the Islamic Conference 2007). Applicants submitted a short form and deposited the Hajj fee at one of 1,559 bank branches across Pakistan between July 20 and August 15, 2005. The remaining 60,000 visas were allocated by private tour operators. As most Pakistanis are not eligible for the special quotas and the private operators are typically more expensive, the lottery is the primary source of visas for most Pakistanis. A total of 134,948 people were part of the government lottery with 59% successful.[4]

The Hajj lottery is conducted over *parties* of up to 20 individuals who will travel and stay together during the pilgrimage.[5] Parties are formed either voluntarily, often along family lines, or by staff of the bank branches. Parties are assigned into separate strata for the two main Islamic sects (Sunni/Shia), eight regional cities of departure, and two types of accommodation that vary slightly in housing quality. A computer algorithm selects parties randomly from each stratum until the quota of individuals for that stratum is full. This process leads to a slightly lower chance of success for larger parties; if the selected party is larger than the remaining quota, it is set aside and another is randomly chosen from the remaining pool.

The lottery selection algorithm was designed and implemented by an independent and reputable third party, and there were no reports of lottery manipulation. The rich and connected typically go through a private Hajj tour operator or the special

3. See Clingingsmith, Khwaja, and Kremer (2008) for a fuller description.

4. Excluding applicants automatically given visas because they applied unsuccessfully in the two preceding years.

5. In our survey sample, 34% of applicants were in a party with fewer than five people (only 1% were in a party of one), and 70% were in a party with ten or fewer people. The remaining 30% were in groups of size eleven to twenty.

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quota rather than participating in the Hajj lottery. Consistent with the hypothesis of random assignment, success in the Hajj lottery is individually and jointly uncorrelated with applicant characteristics listed on the Hajj application forms, such as gender, marital status, year of birth, education, branch of application, and whether applicants listed a telephone number. A joint F-test fails to reject the null hypothesis of random assignment with a p-value of .98 (Table I, Panel A). With observations on 134,948 Hajj applicants, if the lottery were subject to influence, one would expect significant differences by characteristics such as education level, so this offers a pretty strong test.

Among the successful Hajj applicants we surveyed, 99% went on the Hajj. Some unsuccessful lottery applicants secure a place with a private Hajj operator or through the special quota. Thus, 11% of those who were unsuccessful in the government lottery still performed the Hajj that year.

Because compliance with the lottery is not perfect, we use success in the lottery as an instrumental variable to estimate the effect of performing the Hajj. This yields the local average treatment effect (LATE) for those for whom the outcome of the lottery determines Hajj participation. Our estimation equation is

$$Y_i^k = \alpha^k + \beta^k \,\mathrm{Hajj}_i + \lambda_c + \varepsilon_i^k, \tag{1}$$

where Hajj_i is an indicator variable for whether individual i performed the Hajj, Y_i^k is the kth outcome of interest, and Hajj_i is instrumented by the individual's lottery status. As long as success in the Hajj lottery only affects outcomes by inducing applicants to undertake the Hajj, this provides unbiased estimates of β^k. Although we have no explicit way of ruling out a direct lottery effect, it seems unlikely, because our survey period was 8 to 11 months after the lottery. Potential direct effects, such as disappointment at not receiving a visa, are likely to be short-lived, especially given that individuals reapply. Moreover, such an effect would likely have led unsuccessful individuals to report greater distress, whereas we will show evidence to the contrary.

Equation (1) also includes stratum-by-party size cell fixed effects λ_c, as the randomization was done within strata and there were slightly different chances of success depending on party size. However, because quotas for each departure city were proportional to the number of applications and the chance of success varied only slightly with party size, results are similar without

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TABLE I
RANDOMIZATION CHECKS

	Panel A	Panel B	
	Success in lottery	Success in lottery among interviewed	Success in lottery among interviewed, restricted subsample
Applicant characteristic	Coefficient (SE)	Coefficient (SE)	Coefficient (SE)
Female	−0.001	−0.017	−0.026
	(0.004)	(0.022)	(0.024)
Application number[a]	0.001	−0.003	−0.015
	(0.003)	(0.015)	(0.016)
Travel party number[a]	0.005	0.071	0.037
	(0.006)	(0.068)	(0.072)
Year of birth	0.000	0.001	0.001
	(0.000)	(0.001)	(0.001)
Married	0.009	−0.017	0.006
	(0.008)	(0.063)	(0.073)
Middle school	−0.001	−0.019	−0.012
	(0.005)	(0.037)	(0.041)
High school	0.000	−0.045	−0.050
	(0.006)	(0.046)	(0.051)
Intercollege and up	0.002	−0.005	−0.006
	(0.008)	(0.052)	(0.060)
Branch of application[a]	0.005	−0.004	0.000
	(0.009)	(0.042)	(0.000)
Provided phone number	−0.001	0.080	0.094
	(0.011)	(0.060)	(0.064)
Constant	1.142	−1.464	−2.481
	(0.264)	(2.499)	(2.689)
Observations	—	1,605	1,295
R^2	.02	.06	.10
Joint F-test of individual characteristics (p-value)	.98	.89	.81

Notes. Robust standard errors in parentheses clustered at the party level. Regressions include dummies for place of departure × accommodation category × party size category.

[a]Application number is in units of 100,000; travel party number is in units of 10,000; branch code is in units of 1,000.

*Significant at 10%.
**Significant at 5%.
***Significant at 1%.

the cell dummies. Standard errors are clustered at the party level, because outcomes for people traveling together may not be independent.

Both for ease of exposition and to lessen data-mining concerns, we present our results using thematic indices that are constructed by grouping related questions. For example, for views on

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female education, we construct an index that combines questions about whether girls should receive education, what level of schooling girls should receive, etc. Although results on any component question could potentially be due to chance (Type I error), this is less likely when one simultaneously considers several related questions in an index. Moreover, the use of indices reduces the risk of low statistical power (Type II error). We compute the average effect size (AES) across outcomes (components) within an index following O'Brien (1984) and Kling et al. (2004).[6] For a family of J related outcomes Y^j in an index, with Hajj local average treatment effects π_j, the average effect size is $\tau = \frac{1}{J}\sum_{j=1}^{J}\frac{\pi_j}{\sigma_j}$, where σ_j is the standard deviation of outcome j in the comparison group.[7]

We report standard tests of the null hypothesis of no effect for each individual index. However, because we have 25 indices, we also show that our results are robust to multiple hypothesis testing by using a conservative Bonferroni–Holm test, which makes no assumptions about the correlation of hypotheses. We can reject the null hypothesis of one or more false positives with an α of 0.07. Less conservative methods, such as specifying an acceptable false discovery rate, would make this result even stronger.

III.B. The Survey

We surveyed successful and unsuccessful applicants to the 2006 Hajj lottery five to eight months after the Hajj.[8] The survey includes questions on religious knowledge and practice, tolerance, views on gender, social interaction and roles, political involvement and beliefs, physical and mental health, and business and employment, as well as background information on the household and its members.

6. Results are similar with indices that average over component questions (see the working paper version).

7. To test for τ against the null hypothesis of no average effect, we account for the covariance between the effects π_j by jointly estimating the π_j in a seemingly unrelated regression framework. We stack the J outcomes and use our treatment effects regression fully interacted with dummy variables for each outcome as the right-hand side. The coefficients π_j are the same as those estimated in the outcome-by-outcome regressions. Our stacked regression now gives us the correct covariance matrix to form a test of τ.

8. Conducting a baseline survey was infeasible because the lottery took place less than a month after applicant data were available. Surveys after the lottery would not constitute a valid baseline because the successful applicants were preparing to leave and differentially affected.

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TABLE II
SUMMARY STATISTICS

Characteristic	Adult Pakistani population (restricted > 20 years old)		Full sample		Restricted subsample	
	Mean	Std. dev.	Mean	Std. dev.	Mean	Std. dev.
Age	40.16	16.244	54.575	13.240	55.039	13.246
Female	0.499	0.500	0.490	0.500	0.496	0.500
Married	0.703	0.497	0.943	0.232	0.948	0.222
Illiterate	0.482	0.458	0.402	0.490	0.417	0.493
Intercollege and up	0.201	0.43	0.178	0.383	0.157	0.364
City[a]			0.400	0.490	0.372	0.483
Periurban/ large village[a]			0.274	0.460	0.293	0.455
Rural[a]			0.325	0.470	0.335	0.472
Ballot success			0.533	0.499	0.524	0.500
Monthly expenditures (log)	8.678	0.641	8.832	0.783	8.896	0.726

Notes. $N = 1{,}605$ for full sample, $N = 1{,}295$ for subsample, and $N = 29{,}995$ for adult Pakistani population. The Pakistani adult population is from the MICS 2003–4 survey (restricted to the same districts as in our sample).
[a]City, periurban, and rural classifications comparable to our survey data are not available in the MICS.

The initial sampling frame was the list of all Hajj lottery applicants obtained from the Ministry. The survey area was limited for logistic ease to nine administrative districts in the Punjab province.[9] Surveyors used addresses and telephone numbers provided in the applications to locate applicants and interview them at their residences. The sample was also restricted to Sunni applicants, because there were too few Shia applicants for meaningful inferences to be drawn. To maximize statistical power, we randomly selected equal numbers of winning and losing parties. Within each party, we randomly selected an individual to interview, and, if other party members of opposite gender were identified as living with the individual, we also selected a second person of the opposite gender.

Surveyed applicants are broadly representative of the adult Pakistan population (Table II) with some truncation of the extremes of the socioeconomic distribution, because the poorest cannot afford to go on the Hajj and the rich typically travel on private schemes. Hajj applicants have average education and household

9. The districts were Attock, Islamabad, Rawalpindi, Jhelum, Chakwal, Faisalabad, Sargodha, Multan, and Gujrat.

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TABLE III
SURVEY COMPLETION STATISTICS

	Panel A: Full sample			Panel B: Restricted subsample		
		Lottery status			Lottery status	
Characteristic	Total	Successful	Unsuccessful	Total	Successful	Unsuccessful
Selected for interview	2,537	1,286	1,251	1,995	1,032	963
Raw completed interviews	1,605	855	750	1,295	679	616
Completion rate (%)	63.3	66.5	60.0	64.9	65.8	64.0
Not completed (%)						
Dead/ill	2.1	2.1	2.1	2.3	2.2	2.3
Lives elsewhere	10.4	10.0	10.8	9.8	9.8	9.8
Not found	8.3	6.4	10.3	7.7	6.6	8.9
Not home	7.9	8.7	7.2	8.2	8.7	7.6
Refused	7.9	6.3	9.6	7.2	6.9	7.5

Notes. Interview completion percentages from surveyor reports.

expenditures similar to those for the general population, but are older and more likely to be married. Forty percent are from cities, fairly similar to the general population.

Surveyors completed interviews with 1,605 applicants, 63% of the 2,537 they attempted to interview (Table III, Panel A). However, only 7.9% of the attempted interviews were refused. In about three-quarters of unsuccessful attempts, surveyors were unable to contact or locate applicants. Some applicants lived in a different (out-of-sample) district from the one provided in their application address (often a relative's address they wished to travel with), and it was not logistically possible to survey them. In other cases, addresses were incomplete or incorrect or the applicant was not at home despite three separate attempts. Among applicants the survey team could contact (i.e., interviewed plus refusals), the survey completion rate was therefore 88.8%.

Successful applicants completed the survey at a 66.5% rate, higher than the 60.0% rate for unsuccessful applicants. This difference is statistically significant at the 1% level (Table III, Panel A). Hajjis were easier to locate, perhaps because their participation in the Hajj made them better known in their localities. Successful applicants also had a slightly lower refusal rate, possibly because they regarded the survey as being more pertinent for those who had actually performed the Hajj.

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The unbalanced interview completion between successful and unsuccessful lottery applicants could potentially introduce selection and bias our estimate of the Hajj effect.[10] Therefore, we provide three robustness checks against selection concerns. First, Table I, Panel B, shows that for *completed* interviews, lottery success is not individually or jointly correlated with observable applicant characteristics. Second, our results are robust to demographic controls. None of our 25 index results qualitatively change with controls for district, urban or periurban location, and individual characteristics.[11]

Finally, we examine the robustness of our results to a restricted subsample (Table II) that excludes nine out of the 49 *tehsils* (subdistricts) in our survey area that were particularly difficult to survey. This subsample is balanced on survey completion and reasons for noncompletion. It excludes *tehsils* with more than 25 selected applicants (*tehsils* with smaller samples may generate imbalance mechanically) where the completion rate for successful applicants exceeded that for unsuccessful ones by more than 7%. This subsample contains 81% of the total interviews. Although the completion rate was still somewhat higher for successful applicants (65.8% vs. 64.0%), we fail to reject the null hypothesis of an identical completion rate with a p-value of .66. As in the full sample, lottery success in the interviewed subsample is uncorrelated with applicant characteristics (Table I, Panel B). As our results below show, there is no qualitative change in our estimates in the subsample.

IV. Main Results

This section presents our main results on the impact of the Hajj. Sections IV.A–IV.D examine religious behavior and practices, tolerance, gender attitudes, and well-being, respectively. Our

10. A selection effect would imply that the marginal surveyed successful applicant was *less* willing to give an interview (more uncooperative) and *harder* to locate. This is because the initial randomization guarantees that successful and unsuccessful applicants are distributed identically along any attribute. If selection is introduced by, for example, successful applicants gaining incremental visibility from traveling, then the marginal successful applicant found is slightly less well known *ex ante* than the marginal unsuccessful applicant found. However, it is not clear how such potential selection could generate several of our results, such as a shift from localized to global practice or increased tolerance. If anything, one may expect the opposite for the tolerance result, because selection implies that the interviewed successful applicant is marginally less cooperative.

11. We can use additional data such as assets and expenditure from survey data, and the results are robust to these as well. We prefer not to present these as primary controls due to their potential endogeneity.

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TABLE IV
RELIGION

	AES coefficients		
	Base	Controls	Restricted subsample
(1) Regarded as religious	0.238***	0.230***	0.258***
	(0.06)	(0.055)	(0.061)
(2) Global Islamic practice	0.163***	0.166***	0.171***
	(0.030)	(0.029)	(0.033)
(3) Belief in localized	−0.101***	−0.094***	−0.074**
Muslim practices	(0.032)	(0.031)	(0.035)
(4) Participation in localized	−0.097**	−0.097**	−0.085*
Muslim practices	(0.046)	(0.045)	(0.052)

Notes. Columns give AES estimates for our base, control, and restricted subsample specifications. The AES averages the normalized treatment effects obtained from a seemingly unrelated regression in which each dependent variable is a question in the index. All regressions include dummies for place of departure × accommodation category × party size category, as well as dummies for each of the nine districts in the survey. All results come from IV regressions where the instrument is success in the Hajj lottery. Standard errors in parentheses clustered at the party level: Index component questions with number of components indicated in parentheses: Index 1 (1): Do others regard you as religious? Index 2 (10): How frequently do you: pray, do *tasbih* after prayer, pray in the mosque? Did you pray in the mosque last Sunday? Do you pray optional night prayers? Can you read the Qu'ran? How frequently do you: read the Qu'ran? discuss religious matters? keep fast during Ramadan? keep fast outside Ramadan? Index 3 (10): What is your general view of holy men? Do you regard: visiting holy men as correct? visiting shrines? using amulets? doing a forty-day death ceremony? participating in *maulad mehfil* (special religious gathering)? Do you believe that: a cap is required for prayer? that dowry is mandatory? that widows have different priority in remarriage? that there can be intercession on Judgment Day? Index 4 (4): Do you actively visit holy men? visit shrines? use amulets? participate in *maulad mehfil*?

*significant at 10%; **significant at 5%; ***significant at 1%.

power to detect interaction effects is limited, so we generally do not present interaction results, except in cases where we have strong priors and reasonable power and consistency, as in the case of gender. Our estimates capture the effect of the Hajj five to eight months after pilgrims return. Although this limits our ability to explore persistence of the effects, we do not find any significant changes over the survey period.

The rows of Tables IV–VIII present the average effect size (AES) estimates for each index, including the control and restricted subsample specifications. Because the results are very similar, we focus on the base specification. The component questions in each index are described in the notes to the tables. Table IX further presents results for several index component questions of individual interest. A supplemental Online Appendix presents the Hajj impact estimates and definition details for all the component questions.

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TABLE V
TOLERANCE

	AES coefficients		
	Base	Controls	Restricted subsample
(1) Views of other countries	0.150***	0.147***	0.151***
	(0.04)	(0.04)	(0.04)
(2) Views of other groups	0.131***	0.108**	0.122**
	(0.05)	(0.05)	(0.06)
(3) Harmony	0.128***	0.117***	0.126***
	(0.04)	(0.04)	(0.05)
(4) Peaceful inclination	0.111***	0.121***	0.128***
	(0.03)	(0.03)	(0.04)
(5) Political Islam index	−0.050	−0.044	−0.043
	(0.04)	(0.03)	(0.04)
(6) Views of West	0.029	0.039	0.011
	(0.04)	(0.04)	(0.04)

Notes. See notes to Table IV. Index component questions with number of components indicated in parentheses: Index 1 (6): General view of people from other countries, positive to negative: Saudis, Indonesians, Turks, African, Europeans, Chinese. Index 2 (3): How do members of the following groups compare to your group: different sect? different religion? different ethnicity? Index 3 (4): Do you believe the following groups can live in unity and harmony through compromise over disagreements: sects of Islam? religions? Pakistani ethnic groups? Do you ever pray in a mosque of a different school of thought? Index 4 (8): Belief in incorrectness of: Osama's goals? Osama's methods? How important is peace with India for Pakistan? Should the current India/Pakistan boundary be the permanent border if this leads to peace? Should Pakistan not support/only partly support those fighting the Indian government in Kashmir? How incorrect are: suicide attacks? attacks on civilians in war? physical punishment of someone who dishonors family? Index 5 (5): Agree that: government should enforce Islamic injunctions? religious leaders have right to dispense justice? religious leaders should have direct influence on government? better for politicians/officials to have strong religious beliefs? religious beliefs important in voting for candidate? Index 6 (4): Is it bad for Pakistanis to adopt: Western social values? Western technology? Believe there was Western/Jewish role in 9/11 and 2005 London bombing? Believe West does not take into account interests of countries such as Pakistan?

IV.A. Religious Practices and Beliefs

Hajjis are 13% more likely to report they are regarded as religious persons, a one-fourth standard deviation increase relative to the control group (Table IV, row (1)).

Three indices explore how the Hajj affects religious practice and belief. The first measures global Islamic religious practice, meaning the performance of rites universally acknowledged within the Muslim world. Questions, described in the notes to Table IV, include the applicant's observance of prayer, fasting, and Qur'anic recitation, etc. The Hajj increases the global religious practice index by 0.16 standard deviations (row (2)). This is a fairly large effect, particularly because it reflects practice five to eight months post-Hajj, and not the fervor of a recently returned

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TABLE VI
GENDER

	AES coefficients		
	Base	Controls	Restricted subsample
(1) Views toward women	0.120***	0.116***	0.139***
	(0.04)	(0.04)	(0.04)
(2) Women's quality of life	0.158***	0.138***	0.166***
	(0.05)	(0.05)	(0.06)
(3) Girls' education	0.092**	0.089**	0.097**
	(0.04)	(0.04)	(0.04)
(4) Women in workforce/professions	0.119***	0.112***	0.091**
	(0.04)	(0.04)	(0.04)
(5) Gender authority	−0.005	−0.010	0.005
	(0.02)	(0.02)	(0.03)

Notes. See notes to Table IV. Index component questions with number of components indicated in parentheses: Index 1 (4): How do men and women compare: mentally/intellectually? spiritually? morally/ethically? Are men and women equal? Index 2 (5): Opinion of quality of women's lives in following countries/regions relative to Pakistan: Saudi Arabia, Indonesia/Malaysia, West. Think too many crimes against women in Pakistan: overall? relative to men? Index 3 (5): Should girls attend school? Until what level would permit attendance at coeducational schools for: girls? boys? Until what level should coeducational schools be allowed? How many years should girls study relative to boys? Index 4 (3): Like daughters/granddaughter to work? Like a professional occupation for daughters/granddaughters? Good employment important for daughter/granddaughter-in-law? Index 5 (7): Women better at managing daily affairs? Wives have equal say in deciding number of children? Is it sometimes correct for: woman to divorce husband? marry against parents wishes? When jobs scarce men should not have more right to one than women? Should daughter have equal inheritence share? Do women count equally to men as witnesses?

pilgrim. The Hajj nearly doubled the rate of regular fasting outside of Ramadan (the obligatory month of fasting) to around 9% and increased praying *Tahajjud* (supererogatory) prayers by two-thirds (Table IX, rows (2) and (3)).

In most Muslim countries, there are a variety of Islamic traditions that are not as universally accepted as the global practices examined above. Some of these are specific to particular countries or regions. The Hajj rituals highlight global practices. Local practices might decline because they compete for time and attention with global practices, or because the Hajj induces a shift in belief.

We find evidence of an absolute shift away from local beliefs and practices. Although most pilgrims initially have moderately high levels of local beliefs, the Hajj leads to a 0.10–standard deviation reduction in an index of localized beliefs that are fairly common in South Asia but not among Muslims globally (Table IV, row (3)). Some practices, such as visiting the tombs of saints and using amulets, have roots in local Sufi traditions. Others reflect

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TABLE VII
WELL-BEING

	Panel A: AES coefficients			Panel B: AES	
	Base	Controls	Restricted subsample	Main effect	Male interaction
(1) Rescaled K6 index	−0.206***	−0.206***	−0.200***	−0.369***	0.326***
	(0.05)	(0.05)	(0.05)	(0.08)	(0.09)
(2) Positive feelings	−0.109**	−0.098**	−0.079	−0.149**	0.079
	(0.05)	(0.04)	(0.05)	(0.07)	(0.08)
(3) Index of satisfaction with life and finances	−0.010	0.006	0.011	−0.028	0.036
	(0.04)	(0.04)	(0.04)	(0.05)	(0.08)
(4) Self-rated physical health	−0.213***	−0.219***	−0.239***	−0.320***	0.210**
	(0.05)	(0.05)	(0.06)	(0.07)	(0.10)

Notes. See notes for Table IV. In addition, note that Panel A gives AES estimates for our base, control, and restricted subsample specifications, whereas Panel B adds an interaction between Hajj participation and a male variable to the base AES specification. In Panel B, the instruments are success in the Hajj lottery for the main effect and success interacted with male in the interaction specification. Index component questions with number of components indicated in parentheses: Index 1 (6) [rescaled, high value=less distress]: During the past 30 days, how often did you feel: nervous? hopeless? restless or fidgety? so depressed that nothing could cheer you up? everything was an effort? worthless? Index 2 (5): During the past 30 days, how often did you feel: relaxed and peaceful? content? joyous? How much pleasure do you take in life? Altogether, are you very happy/not at all happy (four-point scale)? Index 3 (3): How satisfied with life as a whole are you (ten-point scale)? How much room for improvement in your quality of life? How satisfied are you with finances (ten-point scale)? Index 4 (2): How good is your physical health (four-point scale)? Have you been free of any 7+ day illness/injury in the past year?

local interpretation of Islamic doctrine, such as giving dowry (Islam instead emphasizes *mehr,* where a man commits to pay his wife in case of divorce) and what remarriage priority should be accorded to widows. Whereas South Asian women often lose status when their husbands die and have little prospect of remarriage, in Islam a widow can readily remarry after a short waiting period.

The Hajj similarly reduces an index of localized religious practice, related mainly to the Sufi traditions mentioned above, by 0.10 standard deviations (Table IV, row (4)). As we noted earlier, some have expressed concern about the erosion of local South Asian traditions. We later present evidence suggesting that the Hajj does not produce a shift in favor of a Saudi version of Islam but rather a move toward the global mainstream.

IV.B. Tolerance

We find that Hajjis display more positive views toward other nationalities and social groups, have greater tolerance, and are more peacefully inclined (Table V).

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TABLE VIII
ENGAGEMENT AND EXPOSURE

	AES coefficients		
	Base	Controls	Restricted subsample
(1) Socioeconomic engagement	−0.002	−0.008	0.011
	(0.02)	(0.02)	(0.02)
(2) Engagement in politics	−0.011	−0.010	−0.024
	(0.03)	(0.03)	(0.03)
(3) Formal knowledge of Islam	0.004	0.000	−0.003
	(0.04)	(0.03)	(0.04)
(4) Diversity knowledge	0.146***	0.139***	0.133***
	(0.04)	(0.04)	(0.05)
(5) Gender knowledge	0.125***	0.116***	0.104**
	(0.04)	(0.03)	(0.04)
(6) Global knowledge	0.083**	0.086**	0.072
	(0.04)	(0.03)	(0.05)

Notes. See notes for Table IV. Index component questions with number of components indicated in parentheses: Index 1 (15): How frequently do you visit: people in your town/village? people outside your town/village? How frequently are you visited by: people in your town/village? people from outside your town/village? How many times in the past year have close family/friends sought advice on: family matters? religious matters? business matters? How many times in the past year have more distant family/friends sought advice on: family matters? religious matters? business matters? Are you a member of following kinds of organizations: religious, professional, school? Do you work as: an employee? for yourself? Index 2 (7): Did you vote in last election? How interested in national affairs? Are you member of political party? Are you a member of a political organization? A social organization? How often do you follow national affairs? Do you have an opinion on how politicians are handling national affairs? Index 3 (10): Name as many of the five pillars of Islam as you can. Correct answer to: How many chapters in the Qu'ran? Can you recite favorite verse of the Qu'ran? What is shortest sura of the Qu'ran? What is longest? How many suras are in the the Qu'ran? What is first revealed verse of the Qu'ran? Is method of prayer described in the Qu'ran? What is percentage required to be given as Zakat (charitable tax)? How long must wealth be held for Zakat to be due? Index 4 (3): Correct answers to: How many accepted schools of thought in Sunni Islam? Is a cap required for prayer? Is saying "talak, talak, talak" sufficient for legal divorce? Index 5 (8): Correct answers to: What was name of prophet's first wife? How many wives is a man allowed at once? Can a Muslim man marry a Jewish or Christian woman? Is dowry mandatory? Further: Have you heard of Islamic law relating to adultery? Do you have an opinion about women's lives in: Saudi Arabia? Indonesia/Malaysia? West? Index 6 (6): How many countries share a border with Pakistan? What country has largest percentage Muslim? What percentage of Nigerians are Muslim? What are world's two most populous countries? Who is the Prime Minister of India? Which is further from Pakistan, England or the United States?

The Hajj increases an index of positive views about people from other countries by 0.15 standard deviations or more than 33% (Table V, row (1)). Hajjis update their beliefs most positively about nationalities they are likely to interact with frequently. The largest positive impact (0.32 standard deviations) is on views toward Indonesians (Table IX, row (4)), the largest non-Saudi pilgrim group and the one Hajjis report as observing the most. Hajjis also have a 0.14–standard deviation more positive view of Saudis (Table IX, row (5)). There is no effect on views of Europeans. Hajjis are also significantly more likely to declare that Indonesians

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TABLE IX
SELECTED SURVEY QUESTIONS

	Question	Coding	Coef.	p-value	Comp. mean	Obs.	R^2
(1)	Do you believe others regard you as religious?	1 = Religious, 0 = Not religious	0.100	.000	0.772	1,541	.033
(2)	Do you pray "Tahajjud Namaz"?	1 = Yes (regularly, occasionally), 0 = No (rarely, never)	0.184	.000	0.281	1,605	.047
(3)	How often did you fast outside of Ramadan during the past year?	1 = Several times per month or more, 0 = Once per month or less	0.041	.006	0.049	1,605	.030
(4)	Is your general view of Indonesian people:	2 = Very positive, −2 = Very negative	0.217	.000	0.362	1,583	.055
(5)	Is your general view of Saudi people:	2 = Very positive, −2 = Very negative	0.110	.026	1.034	1,593	.026
(6)	In your opinion, overall how are people of a different religion compared to your people?	0 = Better or worse, 1 = Same	0.084	.004	0.389	1,604	.025
(7)	Do you believe that people of different religions can live in unity & agreement (harmony) in a given society by making agreements over their differences?	1 = Yes, 0 = No	0.063	.074	0.589	1,270	.036
(8)	Do you ever pray in the mosque of a different maslak than your own?	Binary: 1 = Frequently, 0 = Less often/never	0.034	.021	0.049	1,463	.027
(9)	Do you believe the goals for which Osama is fighting are correct?	1 = Not correct at all/slightly incorrect, 0 = Correct/absolutely correct	0.063	.014	0.068	761	.054
(10)	Do you believe the methods Osama uses in fighting are correct?	1 = Absolutely never/almost never correct, 0 = To small extent/some extent/strongly correct	0.051	.112	0.159	761	.063

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TABLE IX
(CONTINUED)

	Question	Coding	Coef.	p-value	Comp. mean	Obs.	R^2
(11)	How important do you believe peace with India is for Pakistan's future?	1 = Important, 0 = Not important	0.044	.016	0.913	1,155	.020
(12)	Please tell me what you think about the correctness of the following: family members physically punishing someone who has dishonored the family	0 = Correct, 1 = Never correct	0.044	.112	0.261	1,459	.033
(13)	In your opinion, how do men and women compare to each other with respect to the following traits: spiritually	0 = Men are better/equal, 1 = Women are better	0.057	.006	0.111	1,497	.034
(14)	What is your opinion about the quality of women's lives in each of the following countries/regions? Indonesia/ Malaysia	1 = Greater than in Pakistan, 0 = Lower than or equal that in Pakistan; Base variables 5 = Very high, 1 = Very low	0.094	.088	0.262	551	.058
(15)	What is your opinion about the quality of women's lives in each of the following countries/regions? Saudi Arabia	1 = Greater than in Pakistan, 0 = Lower than or equal that in Pakistan; Base variables 5 = Very high, 1 = Very low	0.051	.145	0.322	1,180	.048
(16)	What is your opinion about the quality of women's lives in each of the following countries/regions? West	1 = Greater than in Pakistan, 0 = Lower than or equal that in Pakistan; Base variables 5 = Very high, 1 = Very low	0.087	.051	0.186	646	.091
(17)	Do you think there are too many crimes against women in Pakistan? Overall	Binary: 0 = No, 1 = Yes	0.052	.075	0.597	1,605	.045

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TABLE IX
(CONTINUED)

	Question	Coding	Coef.	p-value	Comp. mean	Obs.	R^2
(18)	Do you think there are too many crimes against women in Pakistan? Relative to men	1 = Against women score < against men score, 0 = Against women score ≥ against men score; Base scores 1 = Yes, a lot; 4 = No, not at all	0.053	.052	0.171	1,135	.026
(19)	In your opinion, girls should attend school	Binary: 0 = Disagree, 1 = Agree	0.028	.039	0.933	1,604	.027
(20)	Until what level would you prefer allow/permit girls in your family to attend coeducational schools (boys and girls in the same school)?	0 = Never, 1 = Primary, secondary, or all levels	0.055	.036	0.722	1,550	.035
(21)	Until what level would you prefer allow/permit boys in your family to attend coeducational schools (boys and girls in the same school)?	0 = Never, 1 = Primary, secondary, or all levels	0.059	.024	0.729	1,550	.036
(22)	Would you like for your daughters or female grandchildren to have a career other than caring for the household?	0 = No, 1 = Yes	0.045	.156	0.540	1,605	.029
(23)	How important are the following characteristics in your son's, grandson's wife?: Good employment or business	0 = Not important, 1 = Important	0.054	.073	0.457	1,562	.028

Notes. Rows contain results from individual IV regressions where the instrument is success in the Hajj lottery, and which include include dummies for place of departure × accommodation category × party size category. p-values are corrected for clustering at the party level.

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are the best practitioners of Islam (regression not reported). In follow-up open-ended interviews, Pakistani Hajjis also reported positive interactions with Indonesians. For example, one older female Hajji said, "I had a very good experience with female Hajjis from Indonesia. They would make space for me whenever I was walking if I gestured for them to do so. One of them even gave me Vicks VapoRub when she found out that I had the flu."

The Hajj also increases an index of beliefs that adherents of different sects, ethnicities, and religions are equal by 0.13 standard deviations (Table V, row (2)). In contrast to the views on different nationalities, the largest move toward equal status is for people of a different religion (Table IX, row (6)), who would not be encountered during the Hajj, as they aren't permitted to attend. Hajjis may thus be willing to extend their notions of tolerance beyond the Muslim world.

Similarly, the Hajj increases an intergroup harmony index by 0.13 standard deviations (Table V, row (3)). The index solicits applicants' views on whether people from different ethnic groups, Islamic sects, and religions could live together in harmony in the same society. It also includes a practice-based question about how frequently the respondent prays in a mosque of a different school of thought. The effect is largest for religion, about which the control group has the lowest belief regarding harmony (Table IX, row (7)). The effect on the respondent praying in a mosque of a different school of thought is also large, almost doubling the control group mean of 4.9% (Table IX, row (8)).

We complement the harmony index by exploring the extent to which the Hajj leads to greater inclination to peace. The Hajj increases a peaceful inclination index by 0.11 standard deviations (Table V, row (4)). Examining some of the component questions, we find that the Hajj almost doubles the number of respondents who declare that Osama bin Laden's goals are incorrect, from 6.8% to 13.1%, and increases the fraction declaring his methods incorrect from 16% to 21% (Table IX, rows (9) and (10)).[12] The Hajj increases the belief that peace with India is important from 91% to 96% (Table IX, row (11)). Hajjis are also 17% more likely to say it is never correct to physically punish someone who has dishonored the family (Table IX, row (12)). Although these results

12. Slightly more than half say his goals are correct; one-third say his methods are correct; quite a few do not answer.

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are consistent with becoming more tolerant, it is also possible that the Hajj confers religious legitimacy on individuals that allows them to more willingly express previously held views.

One might suspect that more orthodox religious practice could be associated with support for political Islam, which advocates a closer relationship between religion and politics and which is often associated with negative perceptions of the West. We see no increase in belief either in the role of religion in politics or in more negative views of the West. It is nonetheless possible that increased tolerance tempers such desires and perceptions, if they do in fact go along with increased orthodoxy.

The Hajj reduces support for political Islam, although the effect is only weakly significant at 15% (Table V, row (5)). The political Islam index includes questions on how deeply religion should be involved in politics. Although the average respondent is likely to see a role for religion in matters of the state, Hajjis are no more likely to do so in spite of an increased attachment to global Islam. In fact, the Hajj significantly reduces beliefs that the state should enforce religious injunctions and that religious leaders should be able to dispense justice on their own. The Hajj does not lead to any increase in an index of negative attitudes toward the West (Table V, row (6)) that encompasses views on adopting Western social values and technologies and commonly held suspicions toward the West. We can reject a negative effect of one-twentieth of a standard deviation with 95% confidence.

We find no evidence that the Hajj affects the tails of the distribution of attitudes toward political Islam and views toward the West. Moreover, although young people may potentially be more susceptible to intolerance, the six tolerance indices examined don't show any differential Hajj effect for younger pilgrims (regressions not reported). If anything, the harmony index shows a more positive effect for the young.

IV.C. The Hajj and Gender

We noted earlier that the Hajj may provide Pakistani pilgrims with a novel opportunity in which men and women interact, perform rituals as equals, and observe the gender roles of other nationalities. Perhaps on account of this, we find that the Hajj causes a 0.12–standard deviation increase in an index of questions about the status of women relative to men along intellectual, spiritual,

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and moral dimensions (Table VI, row (1)). The effect is largest on the spiritual dimension, with an increase of over 50% (from 11% to 17%) in belief that women are better (Table IX, row (13)). Hajjis are also more likely to believe that, although there are gender differences, women's overall status is equal.

The Hajj also increases an index that captures awareness of women's quality of life issues in Pakistan by 0.16 standard deviations (Table VI, row (2)). The index includes respondents' ratings of women's quality of life in other countries relative to Pakistan. The largest effect is on the relative quality of life of Indonesian/Malaysian women being higher, paralleling the previous results on views of other nationalities (Table IX, row (14)). Interestingly, Hajjis show a greater increase in their views on the relative quality of life of women in the West compared to Saudi Arabia (Table IX, rows (15) and (16)). In addition, Hajjis are also more likely to think that crimes against women are high, both on an absolute scale and relative to crimes against men (Table IX, rows (17) and (18)).

Do the more favorable assessment of women's qualities and the greater concern regarding their quality of life in Pakistan go along with a changed view of the role women ought to take in society? We construct three indices to explore the areas of girls' education, women's workforce participation and choice of professions, and the willingness to challenge the authority of men relative to women within the household and in social contexts (Table VI, rows (3)–(5)).

The Hajj increases favorable views toward education for girls by about 0.09 standard deviations (Table VI, row (3)). We find positive Hajj effects on all components except equal educational attainment across gender. The Hajj increases the desire that girls attend school from 93% to 96% (Table IX, row (19)). Hajjis are 8% more willing to allow both their boys and girls to attend coeducational schools at all levels (Table IX, rows (20) and (21)).

Further, the Hajj increases an index of questions about women's workforce participation and profession choice by 0.12 standard deviations (Table VI, row (4)). The Hajj has a substantial impact on each index component. For example, the Hajj increases the fraction desiring that their daughters/granddaughters work from 54% to 60% (Table IX, row (22)). Hajjis are also 12% more likely to think it is important that their future daughter-in-law be employed (Table IX, row (23)).

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However, Hajjis' more favorable views of women do not extend to challenging male authority in the household (Table VI, row (5)). The Hajj has little or no impact on an index that includes questions regarding whether the respondent challenges traditional attitudes on women's roles in domestic matters, such as fertility decisions and marrying against parental wishes, and unequal Islamic rules on gender, such as those related to inheritance laws and providing financial witness. This is perhaps unsurprising given the greater authority and responsibility typically accorded to men along several dimensions within Islam.

Nevertheless, the changed perceptions about gender roles do seem to accompany changes in household behavior. The Hajj increased the fraction reporting occasional marital disagreements by 10 percentage points, a large increase relative to the comparison mean of 15% (regression not reported). Because most married couples perform the Hajj together, it is not possible to separate this effect by the respondent's gender (because it reflects both their own and their spouse's Hajj impact).

Although sample size limitations do not readily allow us to examine heterogeneity of the impact of the Hajj, nevertheless we find that only the girls' education index shows a smaller increase for men than women, who in any case already have close to 100% agreement with the view that girls should be educated (regressions not reported). In fact, the Hajj leads to somewhat larger changes in the indices of views on women and quality of life for male Hajjis than for female ones.

IV.D. Well-Being

Hajjis, primarily women, are more likely to report negative feelings that suggest distress, and are less likely to report positive feelings of well-being (Table VII, rows (1), (2), (5), and (6)). This could potentially be due to the changes in Hajjis' beliefs and frame of reference discussed above (which the psychology literature suggests can lead to stress), to financial stress associated with the cost of the Hajj, or to the impact of the Hajj on physical health.

Hajjis report somewhat higher distress, as measured by a version of the K6 screening scale (Kessler et al. 2003).[13] The index aggregates respondents' experience of six negative feelings in the past month, which we rescale so that a higher value represents

13. We should caution that, to our knowledge, the K6 index has not been formally validated for Pakistan.

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less distress. Although applicants had a low level of underlying distress, the Hajj reduces the index by 0.21 standard deviations (Table VII, row (1)). The Hajj also reduces an index of five positive feelings by 0.11 standard deviations (row (2)). In the restricted subsample, the Hajj effect drops slightly to 0.08 standard deviations with a marginal significance of 11%.

The increase in distress falls entirely on women (Table VII, rows (5) and (6)). On both the rescaled K6 index and the positive feelings index, there is no significant effect of the Hajj on men. Increased distress might be due to the stark contrast between the typical Pakistani woman's daily life and the relatively greater equality and integration experienced during the Hajj. The impact of the Hajj on gender attitudes suggests an increased realization that the constraints and restrictions women are accustomed to in Pakistan may not be part of global Islam. The literature in psychology (Crosby 1991; Lantz et al. 2005) suggests that such changes in frame of reference can induce significant stress, although eventually the stress helps deal with the change.

Although the Hajj has a negative impact on a female pilgrim's emotional state, it does not affect overall life satisfaction, either on average, or for women (rows (3) and (7)). We can reject a negative effect on the index of life satisfaction of about one-tenth standard deviation with 95% confidence. Although we cannot rule it out, we do not see much evidence for the hypothesis that the substantial financial expenditure required by the Hajj creates financial stress that accounts for Hajjis' negative feelings. In fact, we can reject the hypothesis that the Hajj has a negative effect of more than one-twelfth of a standard deviation on the individual component question about satisfaction with finances. The Hajj also does not affect monthly household consumption expenditures or a measure of household assets (regressions not reported). Our interviews reveal that most do not consider the pool of savings for the Hajj as fungible; those unable to go keep these Hajj funds in order to reapply in the future.

The Hajj leads to a 0.21–standard deviation reduction in an index of physical health (Table VII, row (4)) that includes self-reported physical health and illness/injury. Although the decrease in self-perceived health could be due to a change in the reference group for Hajjis from local people to those encountered from other countries on the Hajj, it is not clear that this can account for the doubling in reports of serious physical injury or illness.

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The negative physical health effects are also stronger for women (row (8)), suggesting that part of the negative effect of the Hajj on women's feelings of well-being could be explained by poorer physical health.[14] However, the negative point estimates on men's physical health are larger than the effect on the K6 index (0.11 vs. 0.04 standard deviations), suggesting that the two do not exactly co-move. Also, the coefficient on Hajj lottery success in a regression predicting the K6 index is similar whether or not one controls for physical health, providing further suggestive evidence that the channel for Hajj effects on emotional health is not simply through physical health.

V. Potential Channels

Although our methodology does not provide experimental variation that isolates the potential channels through which the Hajj may impact the pilgrim, we can offer some suggestive evidence. We consider both external channels, which operate by changing the environment a Hajji faces upon return, and internal channels, which reflect changes in Hajjis' beliefs and preferences. We argue that the evidence points toward the importance of the internal channel and, within that, to exposure to people of differing nationalities, sects, and gender.

V.A. External Social Environment

Historical accounts suggest that the Hajj confers social prestige and legitimacy (Donnan 1989; Eickelman and Piscatori 1990; Yamba 1995), although some anecdotal evidence suggests that contemporary Hajjis no longer experience this increase in social status (Scupin 1982). A changed social role may bring expectations for the changed behavior and beliefs that are reflected in our results. For example, Hajjis may be expected to be more religious, and may practice more to fulfill that expectation. Alternatively, increased religious legitimacy may allow Hajjis to express longstanding opinions they have not expressed before, such as those opposing Osama Bin Laden.

We find no impact of the Hajj on an index of social status and engagement (Table VIII, row (1)). The fifteen components include the frequency of social visits, the giving of advice, and

14. Negative physical health effects are not larger for older people.

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membership in social organizations.[15] We might further expect social standing to be reflected in awareness of and engagement in political affairs. In fact, we find no impact of the Hajj on a political engagement index (Table VIII, row (2)) that asks about voting, interest in national affairs, political opinion, and membership in political organizations.

It's possible that the Hajj once led to a much greater change in social roles than it does currently, and that the increased rate of participation in the Hajj due to lower travel costs has reduced the social prestige associated with completing the Hajj. In any case, it does not seem likely that changes in the social role upon return can account for the findings in the previous sections.

V.B. Internal State

The Hajj may alter an individual's internal state, changing beliefs and preferences. For example, Hajjis may undergo a change in religious commitment during the pilgrimage that increases orthodoxy in religious practice and leads them to greater tolerance and belief in gender equity consistent with the Qur'an. Alternatively, Hajjis' increased tolerance and changed gender attitudes may reflect their new exposure to people from different countries and sects and to members of the opposite gender outside their family. Although we cannot rule out the religious dimension, we interpret the evidence as pointing more toward the increased exposure to Muslims from around the world.

We find that the Hajj does not increase an index of formal religious knowledge (Table VIII, row (3)) but does increase indices of experiential knowledge about diversity of opinion within Islam, gender within Islam, and the world more broadly (Table VIII, rows (4)–(6)). The changes in experiential knowledge point to the importance of interaction with and observation of other groups.[16] Furthermore, to the extent that a spiritual transformation and change in religious commitment would be accompanied by a desire to acquire greater religious knowledge, these results do not suggest that such a change is a primary driver of the findings.

15. All except two component questions are not significant even at the 20% level (Online Appendix 5). These two show that Hajjis are slightly more likely to have visitors from out of town, and slightly more likely to be self-employed (p-values of .14 each). However, the magnitudes of these effects are small (5% and 3%).

16. Although some of our results could be due to a generic effect of traveling to a different country rather than the experience of the Hajj, it seems unlikely this accounts for all of the results. For example, it is hard to see why a pure travel effect would lead to more positive beliefs about Indonesians.

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The Hajj increases the index measuring knowledge of diversity within Islam by 0.15 standard deviations (Table VIII, row (4)). Index components include questions on how schools of Sunni thought differ, such as whether it is necessary to wear a prayer cap. The index of gender knowledge and awareness, which combines eight questions on gender and marriage in Islam and on having an opinion on women's issues, increases by 0.13 standard deviations (Table VIII, row (5)).[17] Similarly, Hajjis also show an increase of 0.08 standard deviations in the global knowledge index, which reflects general awareness of the world outside Pakistan (Table VIII, row (6)).[18]

Pilgrims who travel in smaller parties, and thus have more opportunity to interact with non-Pakistanis, experience larger gains in the diversity, gender, and global knowledge indices, as well as in positive views of people from other countries. This is consistent with the idea that the exposure channel is important. The coefficients on the interaction between the Hajj and small party size are large and significant at conventional levels for the gender and global knowledge indices and for positive views of other countries. The small-party interaction effects are 0.13, 0.14, and 0.14 standard deviations, respectively, with p-values of .07, .10, and .06. Point estimates for the group size interaction on other tolerance indices also point to a similar story. The interactions are robust to including other demographic controls and their interactions. However, we cannot rule out that unobservable differences between parties of different size are driving the interaction effects.

We would expect Hajj effects to also be larger for those with less prior exposure to situations similar to the Hajj. However, very few respondents had previously traveled outside Pakistan, which limits our power to test this interaction. There are a few robust interactions with literacy and urban residence, though it is unclear if these relate to the prior exposure that is relevant. Although urban applicants do show a smaller decrease in localized beliefs and practices, the literate see larger gains in some of the

17. Four of the eight questions in the gender knowledge and awareness index are about awareness rather than knowledge: whether the respondent has heard of the Islamic law against adultery and whether they have an opinion on women's lives in three different countries. The Hajj has a somewhat smaller, but still significant, effect on a pure gender knowledge index that is constructed without these questions.

18. We should note that this latter effect falls slightly to 0.07 standard deviations and is marginally significant at 12% in the restricted subsample.

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experiential knowledge measures, suggesting that literacy may partly be picking up other factors such as ability to interact with others. Because party size is often assigned by banks, and is thus less likely to be correlated with unobserved factors, we prefer focusing on it as a test of exposure.

Although Hajjis are also exposed to Saudi Arabia and its people, we think this is unlikely to drive our observed effects. Only the move away from localized religious practices seems consistent with a Saudi influence. Saudi Arabia is generally less accepting of other schools of thought and enforces strict gender segregation; Hajj impacts on gender views are more in line with the more liberal attitudes in other Muslim countries.[19]

Our results thus suggest that Hajjis are likely to be influenced by the practices and beliefs of the typical pilgrim that they encounter during the Hajj, with possibly greater salience to those groups that are more visibly different or are regarded as better in some way, such as in their behavior or organization (factors often mentioned in our interviews). Exposure may therefore induce convergence of belief to the Islamic mean. To the extent that this convergence is a significant force, some of our results may differ for pilgrims from other countries.

VI. Conclusions

Our findings show that the Hajj induces a shift from localized beliefs and practices toward global Islamic practice, increases tolerance and peaceful inclinations, and leads to more favorable attitudes toward women. This demonstrates that deep-rooted attitudes such as religious beliefs and views about others can be changed and also challenges the view that Islamic orthodoxy and extremism are necessarily linked. We conclude with some tentative implications of our results on how social institutions help shape individual beliefs and identity and, at a macro level, how they may foster unity within belief systems.

19. A comparison of gender views across questions from the World Values Surveys shows that Saudis indeed have more conservative gender views than Pakistanis, whereas Pakistanis in turn are more conservative than Indonesians. Fully 62% of Saudis believe a university education is more important for men than women, compared to 24% of Pakistanis and 17% of Indonesians. Similarly, 34% of Saudis do not think that both husband and wife should contribute to household income, compared to 30% of Pakistanis and 15% of Indonesians.

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The social psychology literature suggests that social interactions can lead to either positive or negative feelings toward other groups depending on whether the setting is competitive or cooperative. Several features of the Hajj may create a setting in which the interaction among different groups helps build common purpose and identity. It's worth noting that other social institutions also share such features with the Hajj. Consider medical education, police/military basic training, and international peace camps. Like the Hajj, participants in these institutions leave their everyday environments and their restrictions on mixing across certain lines, such as ethnicity and social class, to enter a setting in which they collectively perform similar actions, often physically strenuous ones, which require cooperation from others. Furthermore, participants in all these institutions accentuate their similarity, often by taking on common dress or hairstyle during the experience and a common title afterward.

It also seems likely that the religious element of the Hajj plays a role beyond providing a cooperative setting. For example, Hajjis' changed attitudes on gender appear to be circumscribed by those norms broadly accepted in Islam. Further, it is plausible that the religious context provides the legitimacy that makes it acceptable for adherents to alter their views. If a Pakistani woman observes her Indonesian counterpart engaging equally with her spouse without compromising her piety, she may also consider it permissible to do so. If pilgrims see others praying somewhat differently yet without interference in the holiest of Muslim places, they may reason that some degree of religious diversity is acceptable.

Our results also shed light on why religions often mandate practices that are costly for individual adherents. Although club good models that apply the framework of individual rationality, as in Iannaccone (1992) and Berman (2000), deliver compelling explanations, additional insights can be obtained using an evolutionary framework in which institutions and prescriptions that reinforce and propagate the religion's beliefs and practices are more likely to persist. By moving pilgrims toward the religious mainstream, the Hajj may help Islam overcome an evolutionary hurdle faced by world religions: maintaining unity in the face of the divergence of practices and beliefs through local adaptations. A number of religious institutions, including written holy texts and central authorities, can help overcome this hurdle. Sunni

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Islam lacks a central authority, and so the role of pilgrimage may be particularly important.

However, achieving convergence and maintaining unity likely require that there be limits to how much diversity is allowed. Too diverse a group may make it difficult to find common ground and too much variance in beliefs increases the likelihood that undesirable religious innovations will spread. It is therefore noteworthy that although people of different faiths made the pilgrimage to Mecca in pre-Islamic times (Armstrong 1997), its institutionalization with Islam's emergence was accompanied by restricting it to Muslims and disallowing non-Islamic practices that were once elements of the pilgrimage.

Both the evolutionary and club good perspectives imply that religions with practices that generate positive externalities for other adherents, provided these are socially efficient, are more likely to persist by raising the attractiveness of being an adherent. Historically, undertaking the Hajj may have created positive externalities for other Muslims both through its effect on tolerance and by facilitating economic trade and the diffusion of economic, cultural, and scientific ideas (Bose 2006). Although we find little evidence of individual medium-term gains in socioeconomic status and engagement in our sample, there is clear evidence for a positive externality in the increased tolerance toward others. Of course, given the significant financial and health costs entailed in undertaking the Hajj, individuals still need to be induced to participate. However, this can be done through religious injunctions, sanctions, and rewards. The Hajj is one of the five pillars of Islam and there is the belief that performing it sincerely cleanses one of all sins.

Models of costly religious practices also often argue that these practices signal commitment and screen out those who may free ride on the religious community. Because individuals have already signaled such commitment by applying to the Hajj (less than 1% withdraw), our comparisons between applicants are not influenced by signaling effects. Our results therefore indeed capture a treatment effect. The fact that the Hajj has a direct treatment effect is not surprising, because one would expect that were it only serving a signaling function, it would decline in observance relative to alternate practices that could screen just as well but at a lower cost.

The findings in this paper also pose the question of whether pilgrimages or central gatherings may foster such unity in other

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belief systems, religious or otherwise, and conversely, whether their absence increases susceptibility to schisms. The Kumbh Mela, bringing together millions of Hindus every three years, along with Catholic pilgrimages to Lourdes and Rome, may play such a cohesive role. Nonreligious examples include national political conventions in the United States that may promote party unity and exchange among delegates from different regions. Conversely, the split between Judaism and Christianity occurred shortly after the destruction by the Romans of the Jewish temple in Jerusalem, which was a central gathering place, in the year 70 A.D. One may even conjecture whether the multiplication of Protestant sects would have been muted had there been a central holy site for pilgrimage among Protestants.

Further insights are likely to come from investigating the impact of the Hajj over different durations and on pilgrims from other countries that differ from Pakistan in their attitudes and exposure, and the impact of other pilgrimages. Because several other countries also allocate Hajj visas by lottery, it should be possible to use the same methodology. More generally, one could use similar approaches to examine the impact of other institutions on social identity. For example, one could use draft lotteries (Angrist 1990) to examine the impact of military service on social identity or regression discontinuity designs to examine the impact of professional training on beliefs and attitudes. Building up evidence from a series of such studies would shed additional light on the broader roles played by institutions, religious and nonreligious, in the shaping of beliefs and identity and the evolution of ideologies and belief systems.

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APPENDIX: OUTLINE OF THE HAJJ PILGRIMAGE

Overall time in Saudi Arabia: 40 days							
After arrival	Formal rituals of the Hajj–All pilgrims from all nations participate simultaneously, 5 days						Before departure
		Day 1	Day 2	Day 3	Day 4	Day 5	
Pakistani pilgrims have a staggered arrival up to 35 days before the formal rituals of the Hajj. During this time, they typically perform *umrah* and pray in the Masjid al-Haram in Mecca. If they will not be in the country for long after the formal rituals, they may visit Medina, though this is typically done afterward.	Morning	*Ihram* garments are donned. *Tawaf* and *sa'y* are performed in the Masjid al-Haram in Mecca.	Travel to Mount Arafat at dawn, 25-30 kilometers away.	Travel to Mina from Muzadalifah. Participate in *jamarat* ritual. Pilgrims symbolically "stone the devil" by throwing the pebbles they have collected at three pillars.	Return to Mina if night spent in Mecca.		Pakistani pilgrims have a staggered departure up to 35 days following the formal rituals of the Hajj. During this time, they typically make a trip to Medina for several days, where they pray at the prophet's mosque, the Masjid al-Nabawi. The trip to Medina is common for all nations. Outside this time, they perform *umrah* and pray in the Masjid al-Haram in Mecca.
	Afternoon	Travel to the town of Mina, 5-6 km away. Rest, prayer, and reading of the Qu'ran.	Stay on plain of Arafat until sunset in prayer and contemplation.	Sacrifice an animal. The meat is given to the poor. A voucher may be purchased for this. Pilgrims released from most *ihram* restrictions. They have their heads shaved.	Complete *jamarat* ritual, stoning all three pillars in Mina.	Complete *jamarat* ritual, again stoning all three pillars in Mina. Pilgrims must leave Mina for Mecca by sunset. Otherwise they must repeat the *jamarat* the following day before returning.	
	Night	Spend night in tents at Mina.	After sunset, begin journey to Muzdalifah, about 15 km away. Spend night in the open at Muzadalifah. Pray and collect pebbles for *jamarat* the next day.	Return to Masjid al-Haram to perform *tawaf*. Pilgrims may also do this the morning of the fourth day. Spend night in either Mecca or Mina.	Spend night in Mina.	Spend night in Mecca. Pilgrims perform a farewell *tawaf* before leaving Mecca.	

Tawaf: Four quick circumambulations of the ka'ba followed by three leisurely circumambulations. Pilgrims say a set of prayers as they walk.
Sa'y: Walking seven times back and forth between the hills of Safa and Marwah, now enclosed in the Masjid al-Haram.
Umrah: A minor pilgrimage that can be done at any time, consisting of a *tawaf* and *sa'y* while wearing the *ihram.*

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REFERENCES

Ahmed, Q. A., Y. M. Arabi, and Z. A. Memish, "Health Risks at the Hajj," *The Lancet,* 367 (2006), 1008–1015.
Angrist, Joshua, 1990. "Lifetime Earnings and the Vietnam Era Draft Lottery: Evidence from Social Security Administrative Records," *American Economic Review*, 80 (1990), 313–336.
Armstrong, K., *Jerusalem: One City, Three Faiths* (New York: Ballantine Books, 1997).
Aronson, E., and S. Patnoe, *The Jigsaw Classroom* (New York: Longman, 1997).
Azarya, V., *Aristocrats Facing Change: The Fulbe in Guinea, Nigeria, and Cameroon* (Chicago: University of Chicago Press, 1978).
Barro, Robert, and R. M. McCleary, "Religion and Economic Growth across Countries," *American Sociological Review*, 68 (2003), 760–781.
Berman, Eli, "Sect, Subsidy, and Sacrifice: An Economist's View of Ultra-orthodox Jews," *Quarterly Journal of Economics,* 115 (2000), 905–953.
Bianchi, R., *Guests of God: Pilgrimage and Politics in the Islamic World* (Oxford, UK: Oxford University Press, 2004).
Boisjoly, J., G. Duncan, M. Kremer, D. Levy, and J. Eccles, "Empathy or Antipathy? The Consequences of Racially and Socially Diverse Peers on Attitudes and Behaviors," *American Economic Review*, 96 (2006), 1890–1906.
Bose, Sugata, *A Hundred Horizons: The Indian Ocean in the Age of Global Empire* (Cambridge, MA: Harvard University Press, 2006).
Clingingsmith, David, Asim Ijaz Khwaja, and Michael Kremer, "Estimating the Impact of the Hajj: Religion and Tolerance in Islam's Global Gathering," KSG Working Paper RWP08-22, 2008.
Crosby, F., *Juggling: The Unexpected Advantages of Balancing Career and Home for Women and Their Families* (New York: Free Press, 1991).
DeVries, D. L., and R. E. Slavin, "Teams-Games-Tournaments (TGT): Review of Ten Classroom Experiments," *Journal of Research and Development in Education*, 12 (1978), 28–38.
Donnan, Hastings, "Symbol and Status: The Significance of the Hajj in Pakistan," *Muslim World*, 79 (1989), 205–216.
Eickelman, Dale F., and James Piscatori, "Muslim Travellers: Pilgrimage, Migration, and the Religious Imagination," in *Comparative Studies on Muslim Societies,* Vol. 9, Barbara D. Metcalf, ed. (Berkeley: University of California Press, 1990).
Fisman, Raymond, S. Iyengar, E. Kamenica, and Itamar Simonson, "Racial Preferences in Dating," *Review of Economic Studies*, 75 (2008), 117–132.
Glaeser, E., and S. Glendon, "Incentives, Predestination and Free Will," *Economic Inquiry*, 36 (1998), 429–443.
Guiso, Luigi, Paola Sapienza, and Luigi Zingales, "People's Opium? Religion and Economic Attitudes," *Journal of Monetary Economics*, 50 (2003), 225–282.
Iannaccone, Laurence R., "Sacrifice and Stigma: Reducing Free-Riding in Cults, Communes, and Other Collectives," *Journal of Political Economy*, 100 (1992), 271–291.
Ibn Battuta, *Tuhfat al-Nuzzar fi Ghara'ib al-Amsar* (Beirut: Dar al-Kutub al-'Ilmiyya, 2002 [originally published ca. 1355]).
Johnson, D. W., and R. T. Johnson, 1983. "The Socialization and Achievement Crisis: Are Cooperative Learning Experiences the Solution?" in *Applied Social Psychology Annual*, Vol. 4, L. Bickman, ed. (Beverly Hills, CA: Sage, 1983).
Kessler, R. C., P. R. Barker, L. J. Colpe, J. F. Epstein, J. C. Gfroerer, E. Hiripi, M. J. Howes, S-L. T. Normand, R. W. Manderscheid, E. E. Walters, and A. M. Zaslavsky, "Screening for Serious Mental Illness in the General Population," *Archives of General Psychiatry,* 60 (2003), 184–189.

Downloaded from http://qje.oxfordjournals.org/ at Harvard Library on December 9, 2015

Kling, J. R., J. B. Liebman, L. F. Katz, and L. Sanbonmatsu, "Moving to Opportunity and Tranquility: Neighborhood Effects on Adult Economic Self-Sufficiency and Health from a Randomized Housing Voucher Experiment," Princeton IRS Working Paper 481, 2004.
Lantz, P., J. House, R. Mero, and D. Williams, "Stress, Life Events, and Socioeconomic Disparities in Health: Results from the Americans' Changing Lives Study," *Journal of Health and Social Behavior*, 46 (2005), 274–288.
Low, Michael, *Empire of the Hajj: Pilgrims, Plagues, and Pan-Islam under British Surveillance,* unpublished M.A. thesis, 2007.
McCleary, R. M., "Salvation, Damnation, and Economic Incentives," *Journal of Contemporary Religion*, 22 (2007), 49–74.
Naipaul, V. S., *Among the Believers: An Islamic Journey* (New York: Vintage, 1981).
O'Brien, Peter C., "Procedures for Comparing Samples with Multiple Endpoints," *Biometrics*, 40 (1984) 1079–1087.
Organization of the Islamic Conference, *Report on the Seventeenth Islamic Conference of Foreign Ministers: Resolutions on Legal, Information, and Political Affairs* (http://www.oicoci.org/english/conf/fm/17/17%20icfm-political-en.htm, 2007).
Pettigrew, T. F., and L. R. Tropp, "A Meta-Analytic Test of Intergroup Contact Theory," *Journal of Personality and Social Psychology,* 90 (2006), 751–783.
PEW Forum, "Public Expresses Mixed Views of Islam, Mormonism," in *Pew Forum on Religion & Public Life* (http://pewforum.org/surveys/religionviews07/, 2007).
Putnam, Robert, "American Grace: The Changing Role of Religion in America," Presentation, John F. Kennedy School of Government, Harvard University, Fall 2007 Faculty Research Seminar.
Sacerdote, B., and E. L. Glaeser, "Education and Religion," National Bureau of Economic Research Working Paper 8080, 2001.
Scupin, Raymond, "The Social Significance of the Hajj for Thai Muslims," *Muslim World*, 72 (1982), 25–33.
Sherif, M., O. Harvey, B. White, W. Hood, and Carolyn Sherif, *Intergroup Conflict and Cooperation: The Robber's Cave Experiment* (Norman: University of Oklahoma Press, 1954).
Slavin, R. E., and R. Cooper, 1999. "Improving Intergroup Relations: Lessons Learned from Cooperative Learning Programs," *Journal of Social Issues*, 55 (1999), 647–664
Stephan, W. G., "School Desegregation: An Evaluation of Predictions Made in Brown v. Board of Education," *Psychological Bulletin*, 85 (1978), 217–238.
The Sunday Times, "Terror Watch on Mecca Pilgrims," January 21, 2007.
Tajfel, H., "Experiments in Intergroup Discrimination," *Scientific American*, 223 (1970), 96–102.
Tajfel, H., and J. C. Turner, "The Social Identity Theory of Inter-Group Behavior," in *Psychology of Intergroup Relations,* S. Worchel and L. W. Austin, eds. (Chicago: Nelson-Hall, 1986).
Wolfe, M., *One Thousand Roads to Mecca: Ten Centuries of Travelers Writing about the Muslim Pilgrimage* (New York: Grove Press, 1997).
X, Malcolm, *The Autobiography of Malcolm X*, with Alex Haley (New York: Grove Press, 1965).
Yamba, C., *Permanent Pilgrims: The Role of Pilgrimage in the Lives of West African Muslims in Sudan* (Edinburgh: Edinburgh University Press, 1995).

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